

BALAKARTHIK SENTHILVEL PALANI

Automation & Robotics Engineer
ROS2 · SLAM · PLC · Machine Learning · M.Eng. TH Köln

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PROFESSIONAL SUMMARY

Automation and Robotics Engineer with **3+ years of industry experience** delivering autonomous mobile robot fleets and intelligent control systems for large-scale manufacturing environments. Spearheaded end-to-end deployment of **75+ AMRs across 25+ manufacturing facilities**, integrating ROS2, SLAM, PLC, and custom HMI into production-grade systems. Currently pursuing an **M.Eng. in Automation and IT at TH Köln**, with a focus on autonomous navigation, robot manipulation, and AI-based estimation for industrial applications. Seeking robotics engineering and intelligent automation roles in Germany and across Europe.

TECHNICAL SKILLS

Robotics	ROS2, Nav2, MoveIt2, SLAM, Path Planning, Gazebo, URDF, TF2, Costmap
Industrial	TIA Portal, Siemens PLC (S7), HMI, SCADA, Modbus, Profinet, OPC-UA
ML / AI	PyTorch, TensorFlow, scikit-learn, CNN, RNN, TCN, PINN, Time-Series Forecasting, Sim-to-Real
Programming	Python, C++, SQL, SCL, Structured Text (ST), Ladder Logic (LAD)
Data & Tools	Pandas, NumPy, Matplotlib, TensorBoard, Linux, Docker, Git, GitHub
Languages	English (Fluent) · German (Beginner/A2) · Tamil (Native)

COMPETITION & AWARDS

AI for Industry Challenge — Dexterous Cable Manipulation [Active Participant]
Intrinsic (Google) · Open Robotics · Google DeepMind · NVIDIA Feb 2026 – May 2026 | Global Open Competition

- ▶ Competing in a global robotics challenge to develop a visual-sensory-motor policy for autonomous cable insertion and management in electronics assembly workcells.
- ▶ Developing an AI model using ROS2, Gazebo/Isaac Sim, and the Intrinsic Vision Model (IVM), targeting sim-to-real deployment on a physical robot workcell at Intrinsic HQ, California.
- ▶ Solution evaluated on task success rate, insertion precision, collision safety, and overall cycle-time efficiency, reflecting real production KPIs.

PROFESSIONAL EXPERIENCE

Systems & Solution Engineer — Robotics
Novus Hi-tech Robotics Pvt. Ltd.

Jun 2022 – Sep 2024 | Gurgaon, India

- ▶ Spearheaded end-to-end deployment of **75+ AMRs across 25+ manufacturing facilities**, from SLAM-based navigation and integration to customer handover, eliminating manual transport dependency at each site.
- ▶ Architected and deployed a real-time **fleet and traffic coordination system** for concurrent multi-AMR operation in shared factory environments, eliminating routing conflicts and improving material-flow throughput.
- ▶ Developed ROS-based path-planning pipelines and custom **HMI dashboards** for live robot tracking, access control, and diagnostics, reducing operator intervention time across deployed sites.
- ▶ Led cross-functional teams in system integration, hardware testing, and customer training programmes for 25+ client facilities.

Development Engineer — Autonomous Systems
Motherson Automotive Technologies

Apr 2021 – Jun 2022 | Chennai, India

- ▶ Designed full control architecture for an **autonomous stacker system**, integrating ROS path planning with PLC actuation logic (Modbus/Profinet) and a custom HMI, achieving autonomous pallet retrieval without manual intervention.
- ▶ Integrated PLC and ROS platforms with a **Warehouse Management System (WMS)** for real-time material handling logic and live inventory coordination.
- ▶ Implemented Modbus and Profinet protocols for reliable data exchange between robotic and automation subsystems.

Graduate Intern & Automation Trainee
Motherson Automotive Technologies

Nov 2019 – Apr 2021 | Chennai, India

- ▷ Contributed to panel design, servo control configuration, and structured PLC programming (LAD, ST) for industrial automation systems.
- ▷ Supported multidisciplinary engineering tasks involving sensors, variable-speed drives, and automation components during product development cycles.

EDUCATION

M.Eng. in Automation and IT
TH Köln — University of Applied Sciences

Oct 2024 – Present | Gummersbach, Germany

- ▷ Focus: Intelligent Robotics, Machine Learning, Physics-Informed Modelling, Industrial Automation
- ▷ Active projects: ROS2 Nav2 autonomous navigation, Franka Panda manipulation (MoveIt2), PINN-based vehicle sideslip estimation

B.E. in Mechatronics Engineering
Kongu Engineering College

Jun 2016 – May 2020 | Erode, India

SELECTED PROJECTS

Ackermann RC Car — Autonomous Navigation (ROS2 Nav2)

2025–26

Gazebo · SLAM · Nav2 · Reverse Parking [GitHub]

- ▷ Built a ROS2 Nav2 autonomous RC car migrating from Gazebo simulation to a physical vehicle with real-time SLAM obstacle avoidance and autonomous reverse parking.
- ▷ Tuned path-planning algorithms for Ackermann steering constraints, enabling specialised navigation behaviours beyond default Nav2 support.

Franka Panda Robot Manipulation — Object Detection using YOLO

2026

ROS2 · MoveIt2 · YOLO · Obstacle-Aware Motion Planning [GitHub]

- ▷ Developed a Pick-and-Sort robot with YOLO-based object detection and MoveIt2 for collision-free trajectory and obstacle-aware motion planning using Franka Emika Panda.

Physics-Informed Neural Network — Vehicle Sideslip Estimation

2025–26

PyTorch · PINN · RNN · TCN [GitHub]

- ▷ Developed and benchmarked a PINN against standard NN baselines for vehicle lateral dynamics estimation, using physical constraints to improve generalisation.
- ▷ Applied RNN and TCN correctors to refine lateral velocity predictions from the self-loop PINN model.

Data Analytics & AI Modelling Platform

2024

Python · Scikit-learn · TensorFlow · TensorBoard · CNN [GitHub]

- ▷ Built an end-to-end ML platform for CSV/ZIP input, preprocessing, regression, classification, and image-based AI modelling with TensorBoard and confusion-matrix evaluation.

AMR Fleet — Multi-Robot Traffic Coordination (Industry)

2022–23

ROS · SLAM · Python · SCL · PLC

- ▷ Developed an SLAM-based AMR with PLC safety logic and multi-robot coordination software; built HMI for live tracking, diagnostics, and access control.
- ▷ *Confidential project; no public repository available.*

For complete project details including industry deployments, visit: Portfolio

ADDITIONAL INFORMATION

Availability

Open to full-time positions, Werkstudent (working student), internship, or M.Eng. thesis collaboration.

Work Authorisation

Indian national with a valid German residence permit for study (NRW); EU work eligibility is subject to permit conditions.